Ethan Butz

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Education

University of Florida

Ph.D. in Aerospace Engineering — **Graduation:** Spring 2028

Advisor: Dr. John W. Conklin

Research funded by NASA ESTO

M.S. in Aerospace Engineering — August 2024 to December 2025

B.S. in Mechanical Engineering — July 2018 to May 2023

B.S. in Aerospace Engineering

Professional Experience

Precision Space Systems Laboratory | Graduate Research Assistant — January 2024 to Present

GRATTIS Electronics Testing and Design | January 2024 to January 2025

- Validated noise floor performance of capacitance sensing electronics for electrostatic gravity sensor
- Drafted multiple custom circuit boards for sensor test validation and collaborating lab projects
- Assembled equipment and developed data collection programs for testing flight components

GRATTIS Mechanical Design and Metrology | May 2024 to Present

- Designed hardware and electronics testing equipment for the GRATTIS gravitational reference sensor
- Manufactured critical electrostatic sensor components to micrometer precision in exotic materials
- Acquired and operated a CMM to characterize flight assembly precision to support science objectives

GRATTIS Payload Integration and Mission Management | May 2025 to Present

- Ran finite element simulations of flight components under launch loads to inform mechanical design
- Supported DMIG integration from multiple teams for a full spacecraft thermal and structural analysis
- Developed instrument integration procedures and metrology programs for payload assembly and testing

Design and Manufacturing Lab — May 2021 to August 2023

Mech 3 Technical Engineer | May 2023 to August 2023

- Reviewed and gave feedback on student team designs and part drawings for a capstone project class
- Managed assembly process and manufactured custom parts for final design projects
- Taught lecture on design for manufacturing and proper dimensioning and tolerancing

Lead Teaching Assistant | May 2022 to August 2023

- Developed new course curriculum for Fall 2022 to adapt to new budgetary and scheduling constraints
- Re-structured lab environment using the 5S system to optimize storage and create space for new machines
- Supervised teams of teaching assistants in student lab sections to ensure safe and effective shop operation

Teaching Assistant | May 2021 to December 2021

- Taught students traditional manufacturing techniques on lathes and mills, both manual and CNC
- Led students through learning and applying DFM in concept generation, selection, and implementation
- Collaborated with other teaching assistants and faculty to maintain the UF student machine shop

GE Appliances | Technology Co-Op Intern January 2022 to May 2022

- Adapted aspects of a finalized product design to accommodate a UL2158 revision expected post-launch
- Conducted tests on DC units to evaluate metrics for automated consumer maintenance notifications
- Prototyped and designed over 30 concept iterations to fit a new part within the finalized design constraints

Related Experience

University of Florida's Dean of Students Office | Student Assistant May 2019 to August 2021

- Worked with a team to manage over 4000 medical petitions protected by FERPA and HIPAA law
- Developed crisis management skills while helping students complete their trauma related petitions
- Communicated with students and faculty to troubleshoot and optimize new online portal

Projects

Treadmill Tricycle Design Project (Solidworks)

- Collaborated with a team on designing a tricycle powered by linear motion on a treadmill belt
- Performed analysis on the selection of gears, snap rings, shafts, ball bearings, keys, and keyways
- Created CAD models and assemblies of the design for FEA analysis and motion studies
- Wrote a technical report detailing analysis and summarizing results

Decommissioned Airplane Seat Universal Remote (Arduino IDE, Solidworks)

- Reverse engineered TV remotes to understand, intercept, and log their IR signals using an Arduino circuit
- Constructed a second Arduino circuit capable of emulating the IR signals of multiple remotes from one emitter
- Designed and printed a case and keys to house a custom printed PCB, Arduino, and power source for regular use

LILAC: Lunar Imaging, Low Altitude CubeSat Design Project (Solidworks)

- Collaborated with a team of eight to design a 12U CubeSat that could image landing sites on the moon
- Coordinated teammates in an integration role to accommodate each system within mass and size constraints
- Created CAD models and full assemblies of the CubeSat and performed FEA analysis on structural elements
- Developed the electrical system for the CubeSat to generate and deliver power during the one-month mission

Conference Posters

[1] Z. Forrester, **E. Butz**, C. Perkins, J. Siu, P. Wass, and J. W. Conklin, "Testbed Development for Earth Geodesy Gravitational Reference Sensor Technology", American Geophysical Union, 2024.

[2] **E. Butz,** S. Apple, J. Gleason, Z. Forrester, P. Wass, J. W. Conklin, "Structural Design and Performance Evaluation of the GRATTIS Payload Mounting to Support Sensor Performance", American Geophysical Union, 2025

Skills and Experiences

Technical: MATLAB | Solidworks | PC-DMIS | Arduino | Microsoft Suite | Machining | Rapid Prototyping

Non-Technical: Communication | Leadership | Written and Verbal Communication | Crisis Management

Other: 10 years of experience on live audio-video production teams | 4 years of experience in customer service roles